Polyamine-Induced DNA Phase Separation Length Dependence

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Introduction

Phase Separation
Spontaneous macromolecule separation into a dense phase and dilute phase.[2]

Membraneless compartmentalization in cells can be achieved via Liquid-Liquid phase separation (LLPS).[2]

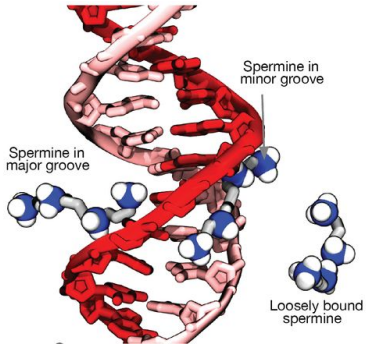
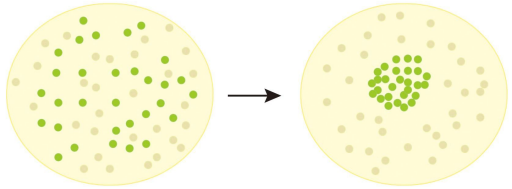
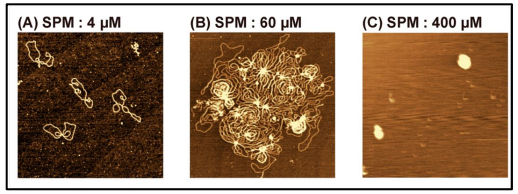


Fig 2. Spermine in the major and minor grooves of the DNA

Liquid-liquid phase separation



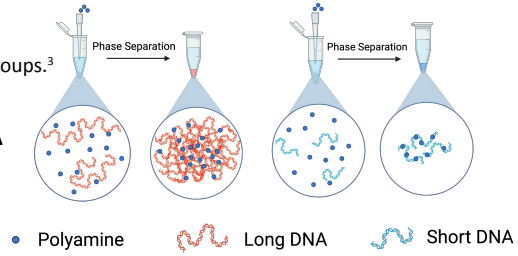


Polyamines induce DNA phase separation, by generating **short-range attraction.**[1]

Objective

DNA phase separation is influenced by nucleotide compositions and methyl groups.[3]

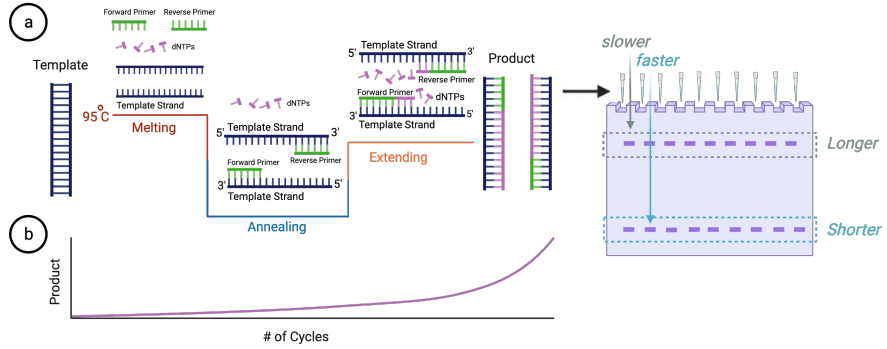
The relationship between DNA size and its effect on phase separation is unknown.



AIM: Engineer DNA fragments spanning a range of sizes which will be subjected to phase separation both in isolation and in combination to explore the correlation between DNA size and its impact on phase separation.

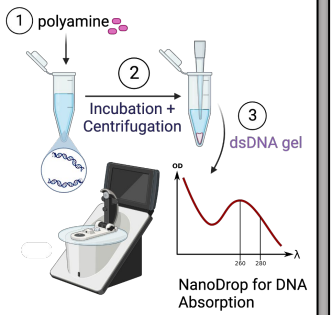
Methods

Polymerase Chain Reaction (PCR)

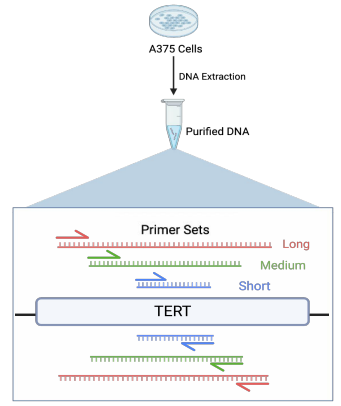


Gel Electrophoresis

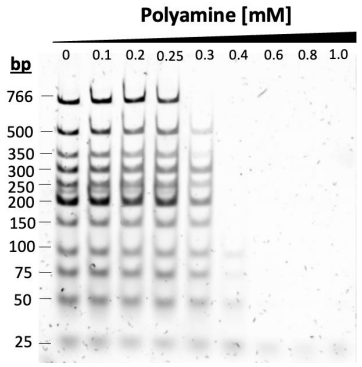
Phase Separation Assay

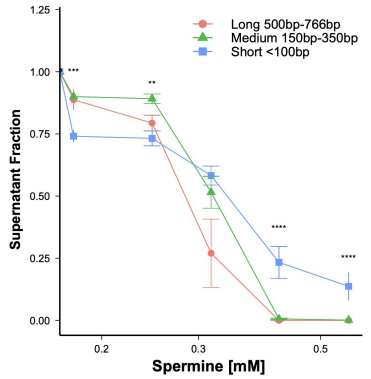


PCR Primer Design



Polyamine-Induced DNA Phase Separation

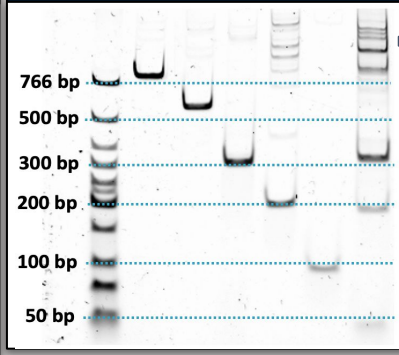




Conclusion/Discussion

Longer DNA fragments have a higher condensation rate than shorter DNA fragments at low polyamine concentrations.
However, possible interactions between short and long fragments could alter phase separation

DNA Fragment Synthesis



Non-specific priming may have arisen due to non-specific binding between the primer and the template, lacking specificity towards the TERT gene.

BLAST results show a possible 300,000 binding sites.

Future Directions

Investigate the difference in phase separation between varying lengths of DNA:
(1) **in isolation**
(2) **in specific combinations**

Acknowledgements

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[1]Raspaud, E., et al. (1998). Precipitation of DNA by polyamines: a polyelectrolyte behavior. *Biophysical journal*

[2]King, J. T., & Shakya, A. (2021). Phase separation of DNA: From past to present. *Biophysical journal*

[3]Hyunju Kang et al. DNA phase separation, *Nucleic Acids Research* 2018